Linux & DevOps Practical Implementation on Ubuntu

Objective:

The objective of this task is to perform and understand essential Linux and DevOps operations on an Ubuntu system. This includes user management, file handling, permissions, process management, package installation, server setup (Apache, MySQL, Tomcat), and text processing using tools like AWK and SED.

Prerequisites:

- Ubuntu 24.04 / 25.04 installed (VM or physical system)
- User with sudo/root privileges
- Internet connection (for package installation)
- Basic knowledge of Linux terminal commands

Implementation Steps:

1) Create a new user:

Ans: Open terminal and Create user and Enter password then successfully created to check id username then enter we get details of new user like id, groups, owners.

```
vboxuser@Arunsai:~$ id techie
uid=1003(techie) gid=1003(techie) groups=1003(techie)
vboxuser@Arunsai:~$ id Arun
uid=1001(Arun) gid=1001(Arun) groups=1001(Arun)
```

2) Navigate to the home directory.

Ans: Open terminal and use the command to change directory to home (cd ~).

```
vboxuser@Arunsai:~$ ~
bash: /home/vboxuser: Is a directory
vboxuser@Arunsai:~$ cd ~
vboxuser@Arunsai:~$
```

3) Create a new directory.

Ans: mkdir directory name to create new directory.

```
vboxuser@Arunsai:~$ mkdir Devops
vboxuser@Arunsai:~$ ls
alphabet  Desktop  Documents  Music     Public      snap       test      Videos
arun.tar  Devops   Downloads  Pictures  sample.war  Templates  test_tar
```

4) List the contents of a directory.

Ans: Use the command (**ls**) to display the contents of the current directory.

```
vboxuser@Arunsai:~$ ls
alphabet  Desktop  Documents  Music     Public      snap       test      Videos
arun.tar  Devops   Downloads  Pictures  sample.war  Templates  test_tar
```

5) Change the current directory.

Ans: Use the command cd directory_name to change the current directory.

```
vboxuser@Arunsai:~$ cd Devops
vboxuser@Arunsai:~/Devops$
```

6) Create a new empty file.

Ans: Use the command touch file_name to create a new empty file.

```
vboxuser@Arunsai:~/Devops$ touch desk
vboxuser@Arunsai:~/Devops$ ls
desk
```

7) View the contents of a file.

Ans: Use the command cat file_name to display the contents of a file.

```
vboxuser@Arunsai:~/Devops$ cat desk
vboxuser@Arunsai:~/Devops$
```

```
vboxuser@Arunsai:~$ cat test
hello world
techie
```

8) Copy a file to another location.

Ans: Use the command cp source_file destination_path to copy a file.

```
vboxuser@Arunsai:~/Devops$ cp desk /tmp/
vboxuser@Arunsai:~/Devops$
```

9) Move a file to another location.

Ans: The file "desk" is successfully found in the /tmp directory using the ls command.

```
vboxuser@Arunsai:~/Devops$ ls
vboxuser@Arunsai:~/Devops$
```

```
vboxuser@Arunsai:~/Devops$ ls /tmp
desk
```

10) Rename a file.

Ans: The file name changes from desk to new desk in the directory.

```
vboxuser@Arunsai:~/Devops$ cd /tmp
vboxuser@Arunsai:/tmp$ mv desk newdesk
vboxuser@Arunsai:/tmp$ ls
newdesk
```

11) Delete a file.

Ans: Use the command rm file_name to delete a file.

```
vboxuser@Arunsai:/tmp$ rm newdesk
vboxuser@Arunsai:/tmp$ ls
snap-private-tmp
```

12) Grant or revoke permissions on a file or directory.

Ans: Use the command chmod 755 Devops to change permissions.

```
vboxuser@Arunsai:~$ chmod 755 Devops
vboxuser@Arunsai:~$ ls -l
total 72
-rw-rw-r-- 1 vboxuser vboxuser    36 Mar 27 10:00 alphabet
-rw-rw-r-- 1 vboxuser vboxuser 10240 Mar 28 07:52 arun.tar
drwxr-xr-x 6 vboxuser vboxuser  4096 Mar 27 06:13 Desktop
drwxr-xr-x 2 vboxuser vboxuser  4096 Mar 29 01:33 Devops
drwxr-xr-x 2 vboxuser vboxuser  4096 Mar 26 15:21 Documents
drwxr-xr-x 2 vboxuser vboxuser  4096 Mar 26 15:21 Downloads
```

13) View the current date and time.

Ans: Use the command date to display the current date and time.

```
vboxuser@Arunsai:~$ date
Sun Mar 29 02:05:09 AM UTC 2026
vboxuser@Arunsai:~$
```

14) Check the system uptime.

Ans: Use the command uptime to check how long the system has been running.

```
vboxuser@Arunsai:~$ uptime
 02:06:28 up 16:27,  1 user,  load average: 0.00, 0.03, 0.00
vboxuser@Arunsai:~$
```

15) View the running processes.

Ans: Use the command ps aux to display all running processes in the system.

```
vboxuser@Arunsai:~$ ps aux
USER         PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root           1  0.0  0.4  23180  8328 ?        Ss   Mar28   0:21 /sbin/init sp
```

16) Kill a running process.

Ans: Use the command kill process_id to terminate a running process or Kill -9 process id.

```
vboxuser@Arunsai:~/Desktop$ ps -ef | grep nginx
root        9582       1  0 07:34 ?        00:00:00 nginx: master process /usr/sbin/nginx -g daemon o
 master_process on;
www-data    9612    9582  0 07:37 ?        00:00:00 nginx: worker process
vboxuser    9621    5077  0 07:37 pts/1    00:00:00 grep --color=auto nginx
vboxuser@Arunsai:~/Desktop$ sudo kill -9 9582
```

17) Install a package using the package manager (e.g., apt or yum).

Ans: Use the command sudo apt install nginx to install a package

```
vboxuser@Arunsai: ~/Desktop
vboxuser@Arunsai:~/Desktop$ sudo apt install nginx
```

```
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
nginx is already the newest version (1.24.0-2ubuntu7.6).
0 upgraded, 0 newly installed, 0 to remove and 41 not upgraded.
vboxuser@Arunsai:~/Desktop$
```

18) Update the system packages.

Ans: Use sudo apt update && sudo apt upgrade -y to update packages.

```
vboxuser@Arunsai:~$ sudo apt update && sudo upgrade -y
```

```
Get:1 file:/cdrom noble InRelease
Ign:1 file:/cdrom noble InRelease
Get:2 file:/cdrom noble Release
Err:2 file:/cdrom noble Release
  File not found - /cdrom/dists/noble/Release (2: No such file or directory)
Hit:3 http://security.ubuntu.com/ubuntu noble-security InRelease
Hit:4 http://in.archive.ubuntu.com/ubuntu noble InRelease
Hit:5 http://in.archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:6 http://in.archive.ubuntu.com/ubuntu noble-backports InRelease
```

19) Create a symbolic link.

Ans: Use the command ln -s target_file link_name to create a symbolic link.

```
vboxuser@Arunsai:~$ ln -s desk link_desk
vboxuser@Arunsai:~$ ls -l
total 72
-rw-rw-r-- 1 vboxuser vboxuser    36 Mar 27 10:00 alphabet
-rw-rw-r-- 1 vboxuser vboxuser 10240 Mar 28 07:52 arun.tar
drwxr-xr-x 6 vboxuser vboxuser  4096 Mar 27 06:13 Desktop
drwxr-xr-x 2 vboxuser vboxuser  4096 Mar 29 01:33 Devops
drwxr-xr-x 2 vboxuser vboxuser  4096 Mar 26 15:21 Documents
drwxr-xr-x 2 vboxuser vboxuser  4096 Mar 26 15:21 Downloads
lrwxrwxrwx 1 vboxuser vboxuser     4 Mar 29 02:55 link_desk -> desk
```

20) Search for files using the find command.

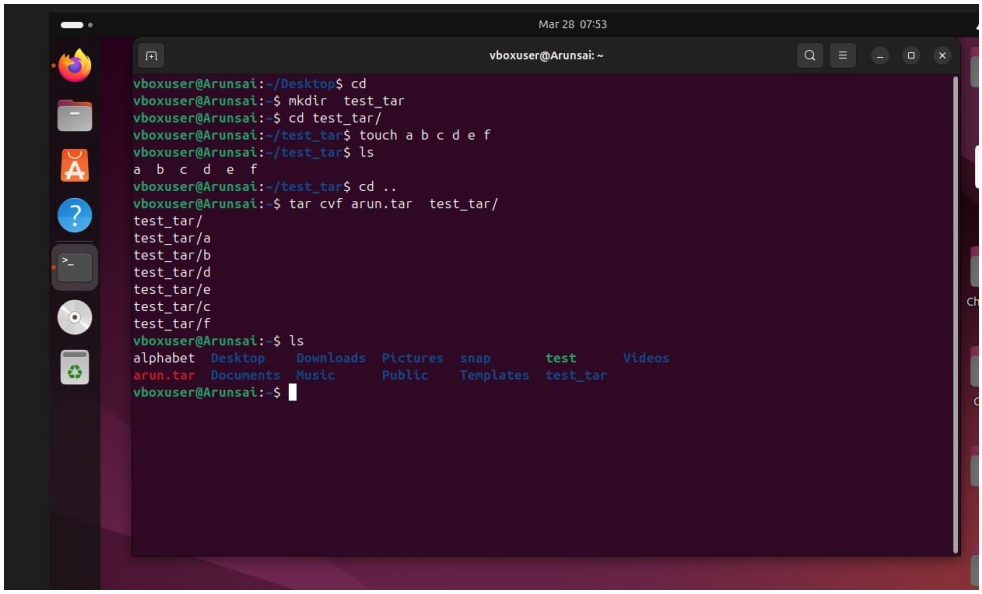
Ans: Use the command find path -name file_name to search for files in a directory.

```
vboxuser@Arunsai:/$ find / -name *.txt
```

```
/etc/brltty/Input/vr/all.txt
/etc/brltty/Input/vs/all.txt
/etc/brltty/Input/tt/all.txt
/etc/brltty/Input/bl/40_m20_m40.txt
/etc/brltty/Input/bl/18.txt
```

21) Compress and decompress files using tar.

Ans**:** Use the command tar -cvf archive_name.tar folder_name to compress the folder.



22) Monitor system resources with top or htop.

Ans: Use the command top to monitor system resources in real-time.

```
vboxuser@Arunsai:/$ top
```

```
  PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM     TIME+ COMMAND
 2521 vboxuser  20   0 3593220 325424  38344 S   1.0  16.2  15:15.45 gnome-shell
```

23) Create and manage user groups.

Ans: Create a group and add a user to the group.

- groupadd devteam
- usermod -aG devteam techie

```
vboxuser@Arunsai:/$ groups techie
techie : techie devteam
vboxuser@Arunsai:/$
```

24) Set up SSH password less authentication.

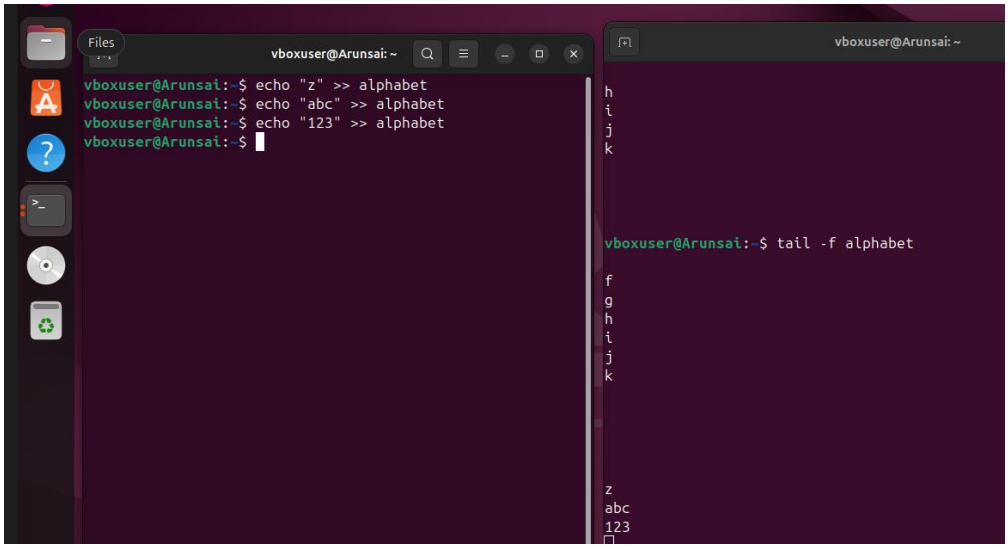
Ans: SSH login works without password.

- Generate SSH key using ssh-keygen.
- Accept default file path.
- Copy public key to remote system using ssh-copy-id.

```
boxuser@Arunsai:/$ ssh-keygen
enerating public/private ed25519 key pair.
nter file in which to save the key (/home/vboxuser/.ssh/id_ed25519):
nter passphrase (empty for no passphrase):
nter same passphrase again:
our identification has been saved in /home/vboxuser/.ssh/id_ed25519
our public key has been saved in /home/vboxuser/.ssh/id_ed25519.pub
he key fingerprint is:
HA256:DuHZ/DZf5Q5AgcIf+by+Ns4l8+wv8x88IooRPVeFiKg vboxuser@Arunsai
he key's randomart image is:
--[ED25519 256]--+
       .. .oo .. |
       .o.+. o.  |
      o  o +..   |
     E =. ..+    |
      +.So ...  .|
       o..o ...o |
       .. +oo.++.|
        o..+=B+oo|
       . . o=++=*|
----[SHA256]-----+
```

25) Monitor log files using tail or grep.

Ans: Use tail -f file_name to monitor file updates in real-time and Modify the file to observe live changes.

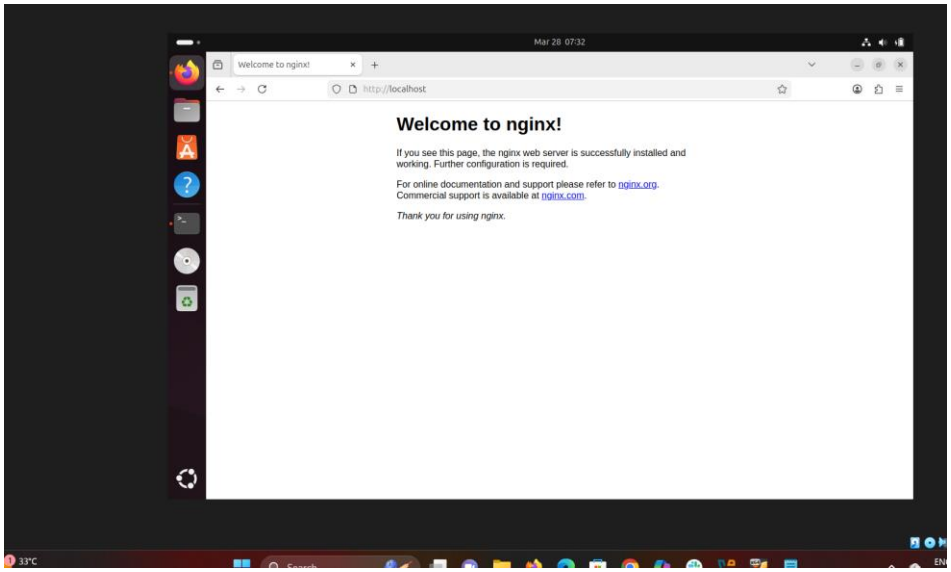


26) Set up a web server (e.g., Apache or Nginx).

1. Install Nginx using package manager.
2. Start and enable the Nginx service.
3. Access the web server using browser

```
Reading state information... Done
nginx is already the newest version (1.24.0-2ubuntu7.6).
0 upgraded, 0 newly installed, 0 to remove and 41 not upgraded.
vboxuser@Arunsai:~/Desktop$ sudo apt install nginx
```

```
root       6622     2  0 06:26 ?        00:00:00 [kworker/0:1H-kblockd]
root       7498     2  0 06:40 ?        00:00:01 [kworker/u4:5-flush-8:0]
root       8002     2  0 06:55 ?        00:00:00 [psimon]
root       8079     2  0 06:55 ?        00:00:00 [kworker/0:0H-kblockd]
root       8186     1  0 07:08 ?        00:00:00 nginx: master process /usr/sbin/nginx -g daemon on; master_process o
www-data   8187  8186  0 07:08 ?        00:00:00 nginx: worker process
root       8195     2  0 07:10 ?        00:00:00 [kworker/0:2-cgroup_free]
root       8225     2  0 07:15 ?        00:00:00 [kworker/u4:1-events_unbound]
root       8232     2  0 07:19 ?        00:00:00 [kworker/u5:2]
root       8243     2  0 07:20 ?        00:00:00 [kworker/0:2H-kblockd]
```



27) Configure and secure a MySQL Database.

Ans: Steps to Secure MySQL Database

- Install MySQL server.
- Start and enable MySQL service.
- Configure security settings.

```
vboxuser@Arunsai:/$ sudo apt install mysql-server
```

```
vboxuser@Arunsai:~$ sudo systemctl start mysql
vboxuser@Arunsai:~$ sudo systemctl enable mysql
Synchronizing state of mysql.service with SysV service script with /usr/lib/syst
emd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable mysql
```

```
vboxuser@Arunsai:~$ sudo systemctl status mysql
● mysql.service - MySQL Community Server
     Loaded: loaded (/usr/lib/systemd/system/mysql.service; enabled; preset: en>
     Active: active (running) since Sun 2026-03-29 05:17:07 UTC; 3min 20s ago
   Main PID: 4438 (mysqld)
     Status: "Server is operational"
      Tasks: 38 (limit: 2261)
     Memory: 353.4M (peak: 367.9M)
        CPU: 4.612s
     CGroup: /system.slice/mysql.service
             └─4438 /usr/sbin/mysqld

Mar 29 05:17:05 Arunsai systemd[1]: Starting mysql.service - MySQL Community Se>
Mar 29 05:17:07 Arunsai systemd[1]: Started mysql.service - MySQL Community Ser>
lines 1-13/13 (END)
```

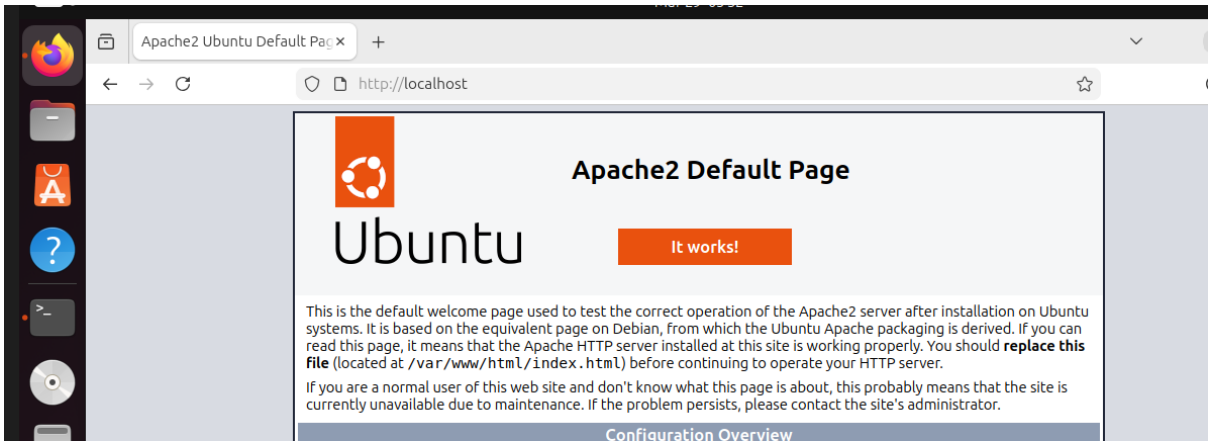
28) Set up a Application Server (e.g.,Apache Tomcat)

Ans: **Steps:**

- Install Apache server.
- Start and enable the service.
- Access the server through browser.

```
vboxuser@Arunsai:~$ sudo systemctl start apache2
vboxuser@Arunsai:~$ sudo systemctl status apache2
● apache2.service - The Apache HTTP Server
     Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: >
     Active: active (running) since Sun 2026-03-29 05:29:05 UTC; 4s ago
       Docs: https://httpd.apache.org/docs/2.4/
    Process: 6118 ExecStart=/usr/sbin/apachectl start (code=exited, status=0/SU>
   Main PID: 6134 (apache2)
      Tasks: 55 (limit: 2261)
     Memory: 5.4M (peak: 5.5M)
        CPU: 103ms
     CGroup: /system.slice/apache2.service
             ├─6134 /usr/sbin/apache2 -k start
             ├─6136 /usr/sbin/apache2 -k start
             └─6137 /usr/sbin/apache2 -k start

Mar 29 05:29:05 Arunsai systemd[1]: Starting apache2.service - The Apache HTTP >
Mar 29 05:29:05 Arunsai apachectl[6133]: AH00558: apache2: Could not reliably d>
Mar 29 05:29:05 Arunsai systemd[1]: Started apache2.service - The Apache HTTP S>
lines 1-17/17 (END)
```



29) create a service file for Apache Tomcat.(Should execute by using systemtctl command)

30) Print specific columns from a delimited file.

Ans: The awk command successfully prints selected columns from the file.
Only specified columns are displayed in the output.

```
vboxuser@Arunsai:~$ cat > test
mahi
abhi
raj
vboxuser@Arunsai:~$ awk '{print $1, $2}' test
mahi
abhi
raj
```

31) Filter and print lines based on a specific pattern or condition.

Ans: 1. Use grep to filter lines based on text pattern.
2. Use awk to filter lines based on conditions.

```
vboxuser@Arunsai:~$ grep "raju" test
raju 60
vboxuser@Arunsai:~$ awk ' $2 > 50' test
mahi 80
raju 60
vboxuser@Arunsai:~$
```

32) Calculate and print the average, sum, or other statistics of a column.

Ans: Use awk to calculate sum and average of a numeric column.

- awk '{sum += $2} END {print "Sum:", sum}' test
- awk '{sum += $2; count += 1} END {print "Average:", sum / count}' test

```
vboxuser@Arunsai:~$ awk '{sum += $2} END {print "Sum:", sum}' test
Sum: 180
vboxuser@Arunsai:~$
```

```
vboxuser@Arunsai:~$ awk '{sum +=$2; count +=1} END {if (count > 0) print "Average:", sum / count}' test
Average: 60
vboxuser@Arunsai:~$
```

33) Perform string manipulation, such as extracting substrings or changing case.

Ans: The sed command successfully replaces specific strings in the file.

```
mahi 80
raju 60
sai 40
vboxuser@Arunsai:~$ sed 's/mahi/MAHI/' test
MAHI 80
raju 60
sai 40
vboxuser@Arunsai:~$ sed 's/raju/RAJU/' test
mahi 80
RAJU 60
sai 40
vboxuser@Arunsai:~$
```

34) Count the occurrences of a specific pattern in a file.

Ans: Use grep -c "pattern" file to count occurrences.

```
vboxuser@Arunsai:~$ cat > test
mahi
raju
sai prasad
raju
vboxuser@Arunsai:~$ cat test
mahi
raju
sai prasad
raju
vboxuser@Arunsai:~$ grep c "raju" test
grep: raju: No such file or directory
vboxuser@Arunsai:~$ grep -c "raju" test
2
vboxuser@Arunsai:~$ grep -c "mahi" test
1
```

35) Sort lines based on a specific field or column.

Ans: Use the sort command with column options to sort data based on a specific field.

```
vboxuser@Arunsai:~$ cat > test
kumar
sai
ganesh
mahi
vboxuser@Arunsai:~$ sort -k1 test
ganesh
kumar
mahi
sai
```

36) Merge multiple files based on a common field or column.

Ans: 1. Create two files with a common column.
2. Sort both files.
3. Use join command to merge them.

```
vboxuser@Arunsai:~$ cat > file1
1 mahi
2 raju
3 sai
vboxuser@Arunsai:~$ cat > file2
1 80
2 40
3 20
vboxuser@Arunsai:~$ join file1 file2
1 mahi 80
2 raju 40
3 sai 20
```

37) Substitute text in a file using search and replace.

Ans: Use the sed command to search and replace text in a file.

```
vboxuser@Arunsai:~$ sed 's/a/A/' test
kumAr
sAi
gAnesh
mAhi
vboxuser@Arunsai:~$
```

38) Delete specific lines based on a pattern or line number.

Ans: Use the sed command to delete lines either by pattern or by line number.

```
vboxuser@Arunsai:~$ cat test
kumar
sai
ganesh
mahi
vboxuser@Arunsai:~$ sed 2d test
kumar
ganesh
mahi
vboxuser@Arunsai:~$
```

39) Append or insert text before or after a specific pattern or line.

Ans: Steps for before and after text pattern

- Use sed '2i text' file to insert before a line.
- Use sed '/pattern/a text' file to append after a pattern.

```
vboxuser@Arunsai:~$ cat test
kumar
sai
ganesh
mahi
vboxuser@Arunsai:~$ sed '2i rajesh' test
kumar
rajesh
sai
ganesh
mahi
vboxuser@Arunsai:~$ sed '/ganesh/a gopi' test
kumar
sai
ganesh
gopi
mahi
```

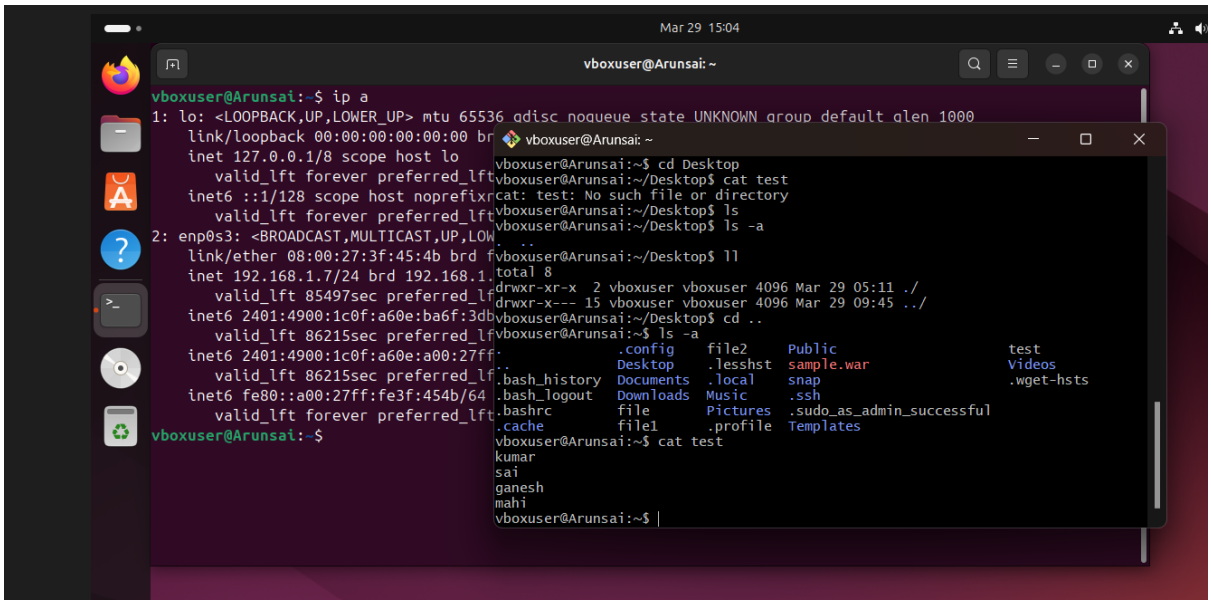
40) Print only specific lines from a file.

Ans: In two ways for specific line to print
Use sed -n 'line_numberp' file to print a specific line.
Use sed -n 'start, endp' file to print a range of lines.

```
vboxuser@Arunsai:~$ cat test
kumar
sai
ganesh
mahi
vboxuser@Arunsai:~$ sed -n '2p' test
sai
```

41) Copy file from linux to windows machine

Ans: The SSH connection is successfully established between linux and Windows vm, and SCP is used to to transfer files.



42) 5 use cases for AWK and 5 use cases for sed

Ans: AWK and SED are powerful text-processing tools used for filtering, formatting, and modifying data in Linux systems.

1. Insert text

- Add text before a line
  sed '2i text' file

```
vboxuser@Arunsai:~$ sed '2i babu' file1
1 mahi
babu
2 raju
3 sai
```

2. Print specific lines

- Display selected lines
  sed -n '2p' file

```
vboxuser@Arunsai:~$ sed -n '2p' file1
2 raju
```

3. Perform calculations

- Calculate sum, average, etc.
  awk '{sum+=$2} END {print sum}' file

```
vboxuser@Arunsai:~$ awk '{sum+=$2} END {print sum}' test
0
```

4. Print specific columns

- Extract particular columns from a file
  awk '{print $1, $2}' file

```
vboxuser@Arunsai:~$ awk '{print $1, $2}' test
kumar
sai
ganesh
mahi
```

5. Perform calculations

- Calculate sum, average, etc.
  awk '{sum+=$2} END {print sum}' file

```
vboxuser@Arunsai:~$ sed '2a babu' file1
1 mahi
2 raju
babu
3 sai
```